Smartlogic™ THE CONTENT INTELLIGENCE COMPANY

White Paper

SKOS vs. OWL; the design decisions for Smartlogic's Semaphore

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In today's enterprise, unstructured information has become one of the most valuable assets of the organization and its rate of growth is staggering. The need to extract information from internal and external sources and integrate it into existing systems for use in key business decisions is critical to successful business outcomes.

Yet without a mechanism to apply context and meaning to information, the enterprise cannot integrate it into their systems. Semaphore, Smartlogic's semantic platform, has been carefully architected to address this problem.

The underlying design principles of Semaphore are a precise combination of W3C standards and semantic best practices built upon Resource Description Framework (RDF), SKOS and SKOS-XL, which allow organizations to unify all enterprise information into a single information source to identify relationships, explore opportunities and answer questions they could not answer before. Semaphore uses high precision NLP and other sophisticated techniques to produce precise, complete and consistent results.

SKOS vs. OWL

We're often asked why we've chosen SKOS (SKOS-XL) instead of OWL in our technology stack. It's not an either/or conversation; SKOS is subset of OWL, so any SKOS implementation is valid OWL - though not complete. Additionally, we augment SKOS with the parts of OWL, which ensure good model design. And the Semantic Web incorporates the complexity of OWL and the practicality of SKOS as they solve different use cases:

***OWL** helps machines understand information - **SKOS** helps humans interact with information*

OWL facilitates greater machine interpretability of Web content than its predecessors (XML, RDF, and RDFS) by supporting a robust vocabulary and formal semantics. OWL's distinguishing feature - expressivity - lets you model extremely complicated and subtle ideas about your data with a rich collection of operators to describe properties and classes. OWL is flexible; all data modeling statements are RDF triples, modifications after the fact can be easily accomplished by modifying the relevant triple.

Yet as with all solutions there are tradeoffs. In order to help machines understand information, allow application developers to integrate models into applications and support expressivity, OWL supports constructs, which are complex and less appealing to knowledge management solutions:

- The expressivity of OWL allows modelers to identify constraints between classes, entities and properties in a very detailed and complex way – however, there is an inverse relationship between expressivity and scalability; as expressivity increases, scalability decreases.
- OWL's expressivity must be described in a very formal way so that applications and by extension machines can interpret. In contrast, SKOS requires a lesser degree of formalism because the expressivity is left to the reader rather than dictated by a machine.
- OWL is a meta-modelling language used to represent classes and their logical description in order to enable sophisticated reasoning, which is appropriate for modelling. In order to generate classification rules, a

semantic network of terms and relationships is required to produce precise and consistent metadata. These fact-level models, which represent the concepts, labels and the relationships between them come from SKOS (SKOS-XL).

For organizations, the decision to use OWL vs. SKOS should be driven by their particular use case. When building systems that require machine-based inference, OWL should be considered. If however, the organization is building knowledge management solutions, which help knowledge workers effectively interact with enterprise information, then SKOS should be considered.

Semaphore and Simple Knowledge Organization System (SKOS & SKOS-XL)

The modeling and classification needs within an organization run along a continuum and vary greatly. There are indeed disciplines that may require extremely detailed models to drive accurate machine-based inference yet the level of complexity associated with creating an OWL model in most organizations is unnecessary or impractical.

In order to provide our customers with a platform that bridges the gap between humans and information we've taken a practical approach to designing our technology stack. We incorporate the tools and standards that allow for the development of robust models that drive and support sophisticated classification strategies.

***SKOS** helps humans interact with information*

The simplicity of SKOS allows organizations to create taxonomies, thesauri, controlled vocabularies and other model types, which can be used to create precise and consistent metadata. For the working taxonomist who wants to improve search, navigation and document management and build knowledge management solutions that improve organizational outcomes, OWL is complicated and does not define the base constructs required to produce a model that will meet those needs. SKOS (SKOS-XL) is the answer, it is the relevant sub-set of OWL that has all the pre-defined constructs necessary for knowledge management and taxonomic structures.

SKOS and SKOS-XL provide a standard way to represent concepts within knowledge organization systems such as controlled vocabularies, taxonomies and thesauri. SKOS is a practical set of modeling constructs, which can be used to create an extensible, distributed information network across the web that:

- Provide multi-language support – SKOS supports internationalized vocabularies that are used in the classification process to increase precision and consistency. For example, a single concept with an American preferred label of "cat" can also have a German preferred label of "katze", a French preferred label of "chat" and an American alternative label of "pussycat" - all of these labels refer to the same concept.

- Is flexible and extensible – SKOS models address variations between different types of vocabularies as well as variations within types of vocabularies.

- Easily combine with other applications and standards – vocabularies can be used with a variety of application/systems such as, databases, faceted navigation and web browsing using a simple web service to link to a published concept or set of concepts.

- Allow taxonomy sharing within and outside of the enterprise:

- Linking concepts across vocabularies allows you to search across disparate data sets with rich synonym sets, or dive into a repository using a particular vocabulary.
- Vocabularies can be reused in combination with other vocabularies (i.e. US postal codes and US congressional districts).

- Improve classification results – SKOS-XL allows us to take labels, which are strings, and give them behaviors by attaching settings, attributes, relationships and metadata.

When an enterprise can take advantage of Linked Data's value, economies of scale, scope and cost effectiveness are realized.

Semaphore and Semantics; Improve patient care and reduce costs

Each year one in three individuals over the age of 75 will experience a fall. Four out of ten falls will result in hospitalization, a long period of recuperation, surgery or death. As the population ages this problem becomes more acute and more expensive.

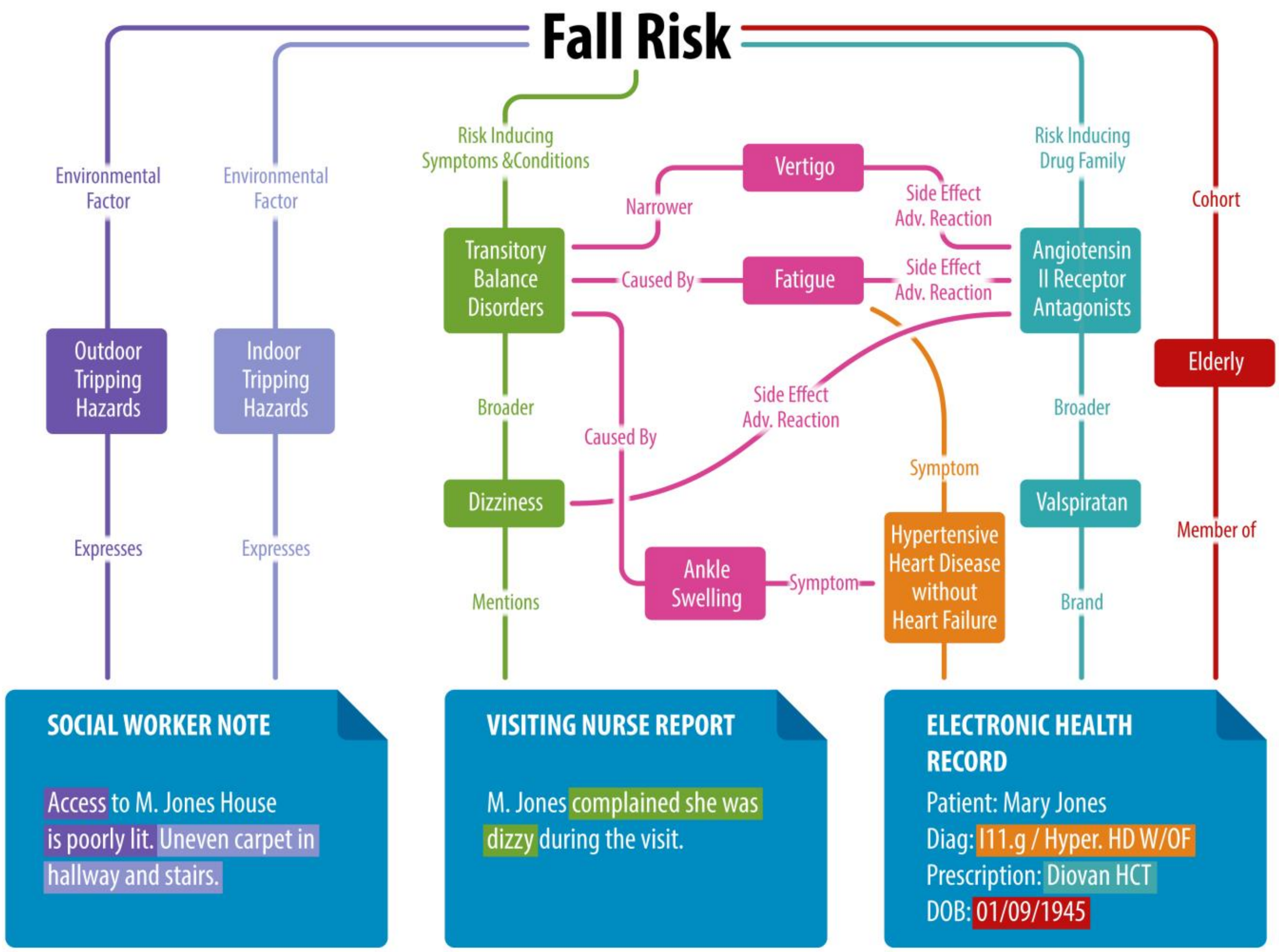
A senior care agency applied the power of Smartlogic's Semaphore and semantics to their disparate data sources to identify individuals who were at risk of falling in order to improve patient care and reduce costs.

The available information such as patient records, at home visiting nurse's notes and social workers observations were stored in different systems within different organizations and varied in structure, document type and nomenclature.

The organization leveraged the U.S. National Library of Medicine's Medical Subject Headings (MeSH) public ontology as a starting point and built out additional detail for their area of interest, adding concepts that corresponded to potential incidents and environmental hazards. They created relationships between drugs and drug families and the risk of someone falling using a specific "risk increasing drugs" relationship. Similarly, they associated environmental hazards identified using Semaphore's Text Miner tool, such as loose carpet, steep stairs and missing handrails with the risk of falling.

This detailed model was used by Semaphore's Classification Server to extract facts from all information assets including doctors' records, social worker reports and comments from care givers and visiting nurses. Sophisticated Natural Language Processing techniques within Semaphore ensured a high degree of accuracy in the created metadata, which was expressed as RDF triples.

Using the SPARQL query language analysts could easily traverse the graph following the arc from concepts within the model to the relevant data points to identify the set of individuals who are likely to fall.



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